Table: Benchmark Probabilistic Scoring & Area Forecast Performance

Model / Protocol Reference	Brier Score (Prob. Error ↓)	BSS vs. Climatology ↑	BSS vs. Persistence ↑	Pinball Loss (Quantile Loss ↓)	Flood-Week MAE (Area Error ↓)	Interval Coverage (Target ~80%)
Task A: LightGBM (Ours)	**0.0650**	**+34.8% (+0.348)**	**+21.7% (+0.217)**	**0.685 km²**	**8.68 km²**	**73.8% (flood) / 92.5% (all)**
Climatology (Calendar Baseline)	0.0996	0.0% (Reference)	-20.1% (-0.201)	1.727 km²	15.52 km²	0.0% (flood) / 71.2% (all)
Persistence (Satellite Baseline)	0.0830	+16.7% (+0.167)	0.0% (Reference)	0.949 km²	8.82 km²	0.5% (flood) / 74.7% (all)

Audit Verified: Strictly adheres to probabilistic protocol scoring rules (Brier Skill Score & continuous quantile pinball loss).
Avoids arbitrary classification thresholds (accuracy/F1). Out-of-fold test pooled across 1,810 county-weeks (374 flood weeks).